Exact Coloring of Claw-Free Graphs: The Role of Structural Representations

[Author names]
Maria Curie-Skłodowska University, Lublin, Poland
[e-mail]

Abstract

For a line graph $G = L(H)$ the chromatic number of G equals the chromatic index of the root graph H. We examine experimentally whether this fact helps when $\chi(G)$ has to be computed exactly. We implemented in Python the recognition of claw-free graphs and line graphs, the reconstruction of a root graph, a DSATUR branch and bound, and three integer programming models solved with HiGHS: an edge model and a maximal-clique model on G, and a star model on the reconstructed root graph that uses Vizing's theorem. The implementation was checked on all 5639 connected claw-free graphs with at most nine vertices. On 1125 line graphs with up to 432 vertices the root model was faster than the edge model on 91% of the instances (median speed-up 1.60). When the reconstruction time is included, it was faster on 71% of the instances (median speed-up 1.23). The gain grows with the density of the root graph and reaches a median of 6.6 for the densest roots. Line graphs of class II roots were much harder for all methods. Among 240 claw-free graphs that are not line graphs, the complements of graphs produced by the random triangle-free process were the hardest instances, although their chromatic number can be found from a maximum matching.

Keywords: claw-free graph, line graph, root graph, graph coloring, edge coloring, DSATUR, integer programming, experimental algorithmics.

1 Introduction

A graph is *claw-free* if it does not contain the star $K_{1,3}$ as an induced subgraph. Line graphs are the best known claw-free graphs. Several problems that are hard in general can be solved in polynomial time on claw-free graphs, for example the maximum independent set problem [13]. Chudnovsky and Seymour developed a structure theory of the class [7], and claw-free graphs are often used to test conjectures on the relation between χ, ω and Δ [10, 28].

Vertex coloring stays hard on this class. For a line graph $G = L(H)$, a proper vertex coloring of G is the same as a proper edge coloring of H, so $\chi(L(H)) = \chi'(H)$. Holyer [17] proved that it is NP-complete to decide whether a cubic graph has chromatic index 3. Hence computing χ is NP-hard already for line graphs.

On the other hand, the root graph gives useful information. By Vizing's theorem [32], $\chi'(H) \in \{\Delta(H), \Delta(H)+1\}$, and by König's theorem [19], $\chi'(H) = \Delta(H)$ for bipartite H. A root graph can be reconstructed in time linear in the size of G [21, 26]. It is therefore natural to ask whether an exact solver should color the vertices of G directly, or first reconstruct H and color its edges. We study this question together with two related ones:

RQ1. How does the difficulty of exact coloring differ between the selected families of line graphs and of claw-free graphs that are not line graphs?

RQ2. Does the root-graph representation of $G = L(H)$ reduce the time needed to compute $\chi(G)$ exactly?

RQ3. Which structural parameters are most strongly associated with the running time of the exact methods?

Our working hypotheses were: (H1) for many line graphs, edge coloring of H is faster than vertex coloring of $L(H)$; (H2) the gain grows with $R = |E(H)|/|V(H)| = |V(L(H))|/|V(H)|$; (H3) the chromatic gap $\chi(G) - \omega(G)$, the degeneracy and the local neighborhood density are more closely related to the difficulty than the number of vertices.

The paper does not propose a new coloring algorithm. Its contribution is an experimental comparison of several formulations of the same problem on the same instances, a set of test families with known properties, and an analysis of how the structure of a graph is related to the cost of exact coloring.

1.1 Related work

Coloring claw-free graphs. Chudnovsky and Seymour [8] proved that $\chi \le 2\omega$ for every connected claw-free graph with $\alpha \ge 3$. Such bounds are studied within the theory of χ-boundedness [28]; for example, $\chi \le \lceil 3\omega/2 \rceil$ holds for graphs without an induced fork and without C_4 [9]. Recently, Vu's conjecture, which bounds χ in terms of the maximum number of common neighbors of two vertices, was proved for claw-free graphs [10]. Our experiments cover two very different parts of the class: line graphs, where $\chi \le \omega + 1$, and complements of triangle-free graphs ($\alpha \le 2$), where the gap $\chi - \omega$ can be large.

Edge coloring. A survey of results up to 2019 is given in [4]. Since then the Goldberg–Seymour conjecture for multigraphs has been proved [6]. For simple graphs the main open question is the Overfull Conjecture: a graph with $\Delta > n/3$ should be of class II exactly when it contains an overfull subgraph with the same maximum degree. It has been confirmed for several families of dense graphs [5]. A $(\Delta + 1)$-edge-coloring can be found in near-linear time [1], and Kowalik [20] gave a quasilinear algorithm for Δ-edge-coloring when $\Delta \ge 2\,\mathrm{mad}(H)$, where $\mathrm{mad}(H)$ is the maximum average degree of a subgraph of H. So the upper bound in Vizing's theorem is cheap to reach, and the hard part of exact edge coloring is to decide the class of the graph. This is what the root model in our study has to do.

Exact vertex coloring. Classical exact methods are DSATUR-based branch and bound [3, 14, 27] and integer programming with the assignment formulation strengthened by clique constraints [22, 23]. Newer approaches use decision diagrams [30], SAT encodings of partial-ordering models [12] and SAT procedures that alternate between cliques and colorings [16]. These methods are usually tested on the DIMACS benchmark and on random graphs. Structured classes such as line graphs, for which the instance has a second representation, are not considered there. We did not find a study that compares, on the same instances, vertex coloring of line graphs with edge coloring of their reconstructed roots.

2 Preliminaries

All graphs are finite and simple. For a graph G we write $n = |V(G)|$, $m = |E(G)|$, $\Delta(G)$ for the maximum degree, $\omega(G)$ for the clique number, $\alpha(G)$ for the independence number, $\chi(G)$ for the chromatic number and $\chi'(G)$ for the chromatic index. The *line graph* $L(H)$ has vertex set $E(H)$, and two vertices are adjacent if the corresponding edges of H share an endpoint; H is a *root graph* of $L(H)$.

Claw-free graphs and line graphs. A graph is claw-free if and only if $\alpha(G[N(v)]) \le 2$ for every vertex v. Every line graph is claw-free, since three pairwise non-adjacent neighbors of a vertex xy of $L(H)$ would be three disjoint edges of H, each meeting $\{x, y\}$. Beineke [2] characterized line graphs by nine forbidden induced subgraphs, one of which is the claw. We use the older characterization of Krausz (see [2, 21]): G is a line graph if and only if its edges can be partitioned into cliques so that every vertex lies in at most two of them. In $L(H)$ such a partition is formed by the *stars* of H, that

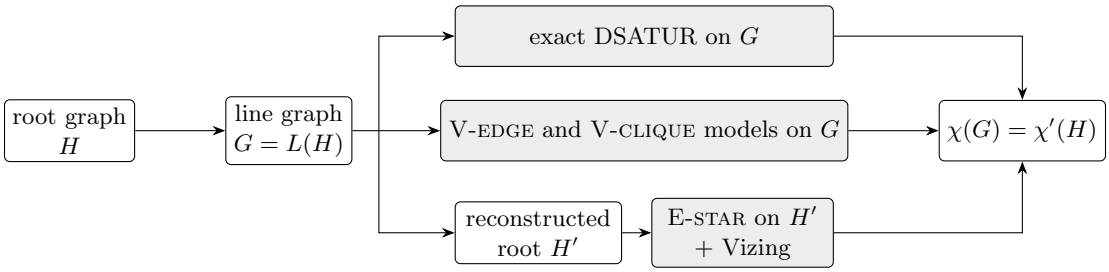


Figure 1: Scheme of the main experiment. The solvers receive only the line graph G with randomly permuted vertices; the root graph used by the star model is reconstructed from G.

is, by the sets of edges incident with a vertex of H. By Whitney's theorem (see [21]), a connected line graph other than K_3 has a unique root graph without isolated vertices. The graph K_3 has two roots, K_3 and $K_{1,3}$; both have chromatic index 3, so the choice does not change the result, although it changes the star model.

Coloring. The stars of H are cliques of $L(H)$, so $\omega(L(H)) \geq \Delta(H)$, and Vizing's theorem gives

$$\Delta(H) \leq \chi'(H) = \chi(L(H)) \leq \Delta(H) + 1 \leq \omega(L(H)) + 1. \tag{1}$$

A graph H is of *class I* if $\chi'(H) = \Delta(H)$ and of *class II* otherwise. Bipartite graphs are of class I [19], and almost all graphs are of class I [11]. A graph H is *overfull* if $|E(H)| > \Delta(H)\lfloor |V(H)|/2 \rfloor$. An overfull graph is of class II, because each color class is a matching and has at most $\lfloor |V(H)|/2 \rfloor$ edges. In particular, a d-regular graph of odd order with $d \geq 1$ is overfull; by the handshake lemma d must then be even.

For general claw-free graphs the gap $\chi - \omega$ can exceed one. Graphs with $\alpha(G) \leq 2$ are exactly the complements of triangle-free graphs, and they are claw-free. If T is triangle-free and $G = \overline{T}$, then a color class of G is a clique of T, i.e. a vertex or an edge. Therefore

$$\chi(\overline{T}) = |V(T)| - \nu(T), \qquad \omega(\overline{T}) = \alpha(T), \tag{2}$$

where $\nu(T)$ is the size of a maximum matching. An optimal coloring is obtained by giving one color to the two ends of each matching edge and a separate color to every other vertex. Both χ and an optimal coloring can thus be found in polynomial time, which gives an independent check of the solvers. If T is bipartite, then $\nu(T) = |V(T)| - \alpha(T)$ by König's theorem and $\chi(\overline{T}) = \omega(\overline{T})$.

3 Algorithms and implementation

The software is written in Python. It uses NetworkX [15] for graphs, SciPy [31] for statistics and as the interface to the HiGHS solver [18], and Matplotlib for plots. The course of the main experiment is shown in Figure 1.

Recognition. Claw-freeness is tested from the definition: for every vertex v and every pair a, b of non-adjacent neighbors of v we look for a neighbor of v adjacent to neither a nor b. A claw found in this way is returned as a certificate. Line graphs are recognized by reconstructing a root graph with the algorithm of Roussopoulos [26] as implemented in NetworkX, applied to each connected component. The result is then checked: the reconstruction gives a map $\varphi\colon V(G) \to E(H')$, and we verify that φ is a bijection and that $uv \in E(G)$ if and only if $\varphi(u)$ and $\varphi(v)$ share an endpoint. For a component equal to K_3 the implementation returns the root $K_{1,3}$. The map φ also turns an edge coloring of H' into a vertex coloring of G.

Structural parameters. For every graph we compute n, m, the density $2m/(n(n-1))$, the minimum, maximum and average degree, the degeneracy, the number of triangles, the clique number (by enumerating maximal cliques) and the mean neighborhood density $D_N(G) = \frac{1}{n}\sum_v D_N(v)$, where $D_N(v) = |E(G[N(v)])|/\binom{d(v)}{2}$ for $d(v) \geq 2$ and $D_N(v) = 0$ otherwise. D_N is the average local clustering coefficient and is used only as a descriptor of local structure. In a claw-free graph the complement of every neighborhood is triangle-free, so by Mantel's theorem the neighborhood of a vertex of large degree has density at least about $1/2$. Hence D_N is high on claw-free graphs.

Exact DSATUR. The first exact method is the DSATUR branch and bound of Brélaz [3]. It selects an uncolored vertex of maximum saturation degree (ties are broken by degree) and tries every admissible color already in use and one new color, as long as the number of colors stays below the best solution found so far. The initial upper bound is given by the greedy DSATUR heuristic and the lower bound is $\omega(G)$; the search stops when the two bounds meet. We record the running time and the number of nodes of the search tree. The node count does not depend on the load of the machine, but it does depend on the branching rule and the bounds used.

Integer programming models. The other methods solve a sequence of k-colorability problems with the assignment formulation [23]. Let I be the set of items to be colored and $\mathcal{Q}$ a family of sets of items whose members must get different colors. With binary variables x_{ic} ($i \in I$, $0 \leq c < k$) we look for a solution of

$$\sum_{c<k} x_{ic} = 1 \quad (i \in I), \qquad \sum_{i \in Q} x_{ic} \leq 1 \quad (Q \in \mathcal{Q},\ c < k). \tag{3}$$

The three models differ only in I and $\mathcal{Q}$:

- V-EDGE: $I = V(G)$ and $\mathcal{Q} = E(G)$; no structural information is used.
- V-CLIQUE: $I = V(G)$ and $\mathcal{Q}$ is the family of all maximal cliques of G, found by the Bron–Kerbosch procedure in NetworkX. The time of this enumeration is included in the running time of the model.
- E-STAR: $I = E(H')$ and $\mathcal{Q}$ is the family of stars of the reconstructed root H', i.e. the Krausz partition of G. This model is applied only to line graphs.

In the vertex models the colors of a maximum clique are fixed to remove the symmetry between colors, and $k = \omega(G), \omega(G)+1, \ldots$ is tested until a coloring is found. In E-STAR the colors of the edges at a vertex of maximum degree are fixed and only $k = \Delta(H')$ is tested; if this model is infeasible, then $\chi'(H') = \Delta(H') + 1$ by (1). For class I roots the model returns an optimal edge coloring, which is mapped back to G. For class II roots it determines the exact value of $\chi(G)$ but does not construct a coloring with $\Delta(H')+1$ colors; such a coloring can be found in near-linear time [1], but this step is not part of the measured procedure, whereas the vertex models always return a coloring. We do not use other shortcuts, such as König's theorem or the overfull test, so that the comparison concerns the models themselves.

An edge model of H with one constraint for every pair of adjacent edges would be identical to V-EDGE. The root graph can therefore help only through the clique (star) constraints and through the bound (1). The star constraints are also much fewer: V-EDGE has $k\sum_{v \in V(H)} \binom{d_H(v)}{2}$ conflict rows and E-STAR at most $k|V(H)|$. The V-CLIQUE model separates the effect of clique constraints, which can be obtained without the root graph, from the effect of the reconstruction and of Vizing's bound.

Every coloring returned by a solver is checked, and the results of all methods on the same graph are compared.

4 Test instances

All instances were generated with fixed random seeds; the generators, the seeds and the resulting graphs are included in the repository given at the end of the paper.

Set A: line graphs. A root graph H is generated, $G = L(H)$ is formed and the vertices of G are randomly permuted. We use five families of roots and 15 graphs for each feasible parameter combination. Erdős–Rényi graphs $G(n, p)$ have $n \in \{10, 20, 30, 40, 50\}$ and $p = \bar{d}/(n-1)$ for the expected average degree $\bar{d} \in \{2, 4, 8, 16\}$, where only combinations with $\bar{d} \le n - 2$ are used; isolated vertices are removed. Random bipartite graphs have parts of sizes $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$, the same n, and edge probability $p = \bar{d}n/(2\lfloor n/2 \rfloor \lceil n/2 \rceil)$ for $\bar{d} < n/2$. Trees are uniformly random labeled trees on the same numbers of vertices. Random d-regular graphs [29] have $d \in \{3, 4, 6, 8\}$ and $n \in \{10, 15, 20, 25, 30, 40, 50\}$, excluding odd nd. Finally, *subdivided regular graphs* are obtained from a random d-regular graph of odd order $n_0 \in \{15, 25, 35\}$, $d \in \{4, 6, 8\}$, by subdividing one edge. The regular graphs of odd order are overfull. The subdivided ones give class II roots that are not overfull:

Observation 1. *Let $d \ge 3$, let F be a d-regular graph of odd order and let H be obtained from F by subdividing one edge. Then H is of class II and is not overfull.*

Proof. Let w be the new vertex. In an edge coloring of H with $d = \Delta(H)$ colors every vertex of degree d meets every color class. A color class is a matching, so it covers an even number of vertices; since $|V(F)|$ is odd, it must also cover w. Then all $d \ge 3$ colors appear at w, which has degree two, a contradiction. Finally, $|E(H)| = d|V(F)|/2 + 1 \le d(|V(F)| + 1)/2 = \Delta(H)\lfloor |V(H)|/2 \rfloor$. □

Although H is not overfull, it contains an overfull subgraph with the same maximum degree, namely $H - w$. This is the kind of obstruction described by the Overfull Conjecture [5].

Set B: claw-free graphs that are not line graphs. By (2), complements of triangle-free graphs are claw-free and their chromatic number is known. We use two families with $n \in \{20, 30, 40, 50\}$ and 30 graphs per value of n. *Co-bipartite* graphs are complements of random balanced bipartite graphs with edge probability 0.3 or 0.6. *Co-triangle-free* graphs are complements of graphs produced by the triangle-free process (pairs of vertices are taken in random order, and an edge is added if it does not close a triangle), run to the end or stopped after half of its edges. None of the 240 graphs is a line graph. This set is a control group, not a sample of all claw-free graphs: its graphs are dense and have $\alpha \le 2$.

Set C: all small claw-free graphs. All connected claw-free graphs with at most nine vertices were generated up to isomorphism by adding one vertex at a time and removing isomorphic copies. This is enough, because every connected graph has a vertex whose removal leaves it connected, and claw-freeness is hereditary. The numbers obtained, $1, 1, 2, 5, 14, 50, 191, 881, 4494$ for $n = 1, \ldots, 9$, and the numbers of line graphs among them, $1, 1, 2, 5, 12, 30, 79, 227, 710$, agree with the sequences A022562 and A003089 of the OEIS [24]. Table 1 summarizes the three sets.

5 Experimental methodology

The experiments were run on one machine (two cores of an Intel Xeon processor at 2.3 GHz, 7 GB of memory, Linux) with Python 3.13.16, NetworkX 3.6.1, NumPy 2.5.3 and SciPy 1.18.1. The `milp` routine of SciPy calls HiGHS with default settings. Two instances were processed at the same time in two separate processes. Each method had a limit of 30 s per graph. For the vertex models this limit covers the whole sequence of k-colorability problems, and each call to the solver receives the remaining time. Times of the models include building the constraint matrix. For a run that reached the limit, we record the measured time, which can exceed the limit slightly (by at most 0.05 s in our runs) because the limit is checked only periodically; the node count of DSATUR is the value at the

Table 1: Test sets; n and m refer to the colored graph G. In set A only feasible parameter combinations were generated.

Set	Family	Graphs	n	m	Mean density	Line graphs
A	Erdős–Rényi	285	4–432	3–7250	0.16	285
A	bipartite	240	4–415	2–6598	0.14	240
A	trees	75	9–49	9–83	0.14	75
A	regular	390	15–200	30–1400	0.15	390
A	subdivided regular	135	31–141	91–981	0.15	135
B	co-bipartite	120	20–50	120–1060	0.77	0
B	co-triangle-free	120	20–50	109–1081	0.78	0
C	line	1067	1–9	0–36	0.45	1067
C	non-line	4572	5–9	7–35	0.62	0

moment of interruption. Medians of times and node counts are taken over all instances, including such runs.

We denote by T_V, T_C and T_E the times of V-EDGE, V-CLIQUE and E-STAR, by T_{rec} the time of the root reconstruction, and define

$$S = \frac{T_V}{T_E}, \qquad S_{\mathrm{tot}} = \frac{T_V}{T_E + T_{\mathrm{rec}}}.$$

S compares the two models, and S_{tot} also charges the root model with the cost of reconstruction. The time needed to compute $\omega(G)$ is common to DSATUR and the vertex models and is not included in T_V and T_C; this slightly favors the vertex models.

Groups are described by medians and, for ratios, by geometric means. Paired samples are compared with the Wilcoxon signed-rank test, independent samples with the Mann–Whitney test, and associations are measured by Spearman's rank correlation ρ. Graphs generated with the same parameters are not independent in the sense of a random sample from all claw-free graphs, so the tests describe the generated families and should not be read as statements about the whole class. Many times are of the order of milliseconds and are affected by measurement noise; for this reason we rely on medians and rank statistics.

6 Results

Table 2: Structural parameters (means unless stated otherwise). For set B the chromatic number is taken from (2).

Set	Family	ω	χ	$\chi > \omega$ [%]	$\max(\chi - \omega)$	$\max \chi/\omega$	Degen.	D_N
A	Erdős–Rényi	11.2	11.2	0.0	0	1.00	11.0	0.50
A	bipartite	9.2	9.2	0.0	0	1.00	9.1	0.46
A	trees	4.2	4.2	0.0	0	1.00	3.2	0.40
A	regular	5.4	5.7	23.3	1	1.33	8.8	0.44
A	subdivided regular	6.0	7.0	100.0	1	1.25	9.0	0.46
B	co-bipartite	17.6	17.6	0.0	0	1.00	22.5	0.80
B	co-triangle-free	13.3	17.5	94.2	11	1.79	24.1	0.80
C	line	3.9	4.0	0.7	1	1.50	3.0	0.54
C	non-line	4.7	4.8	8.7	1	1.33	4.0	0.67

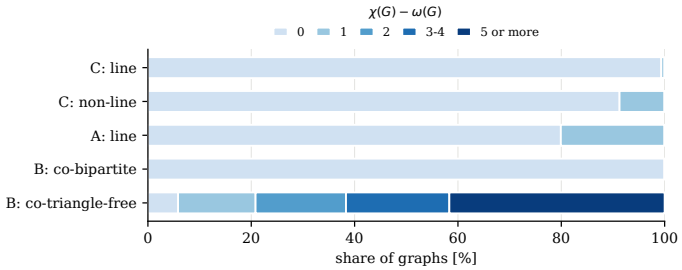


Figure 2: Distribution of the chromatic gap $\chi(G) - \omega(G)$.

6.1 Validation

On set C, DSATUR and V-EDGE solved all 5639 graphs and returned the same chromatic number. On the 1067 line graphs of this set, V-CLIQUE and E-STAR gave the same values as well, which confirms $\chi(G) = \chi'(H')$ for the reconstructed roots. On set B, the values found by DSATUR (180 solved graphs) and by V-EDGE (216 solved graphs) always agreed with $|V(T)| - \nu(T)$ from (2). On set A all methods agreed on every instance solved by more than one of them. We found no inconsistent result.

6.2 Structural properties

Table 2 and Figure 2 describe the instances. Every connected claw-free graph with at most nine vertices satisfies $\chi \leq \omega + 1$. The gap is positive for 406 of them, and only 7 of these are line graphs (the line graphs of class II roots, e.g. $C_5 = L(C_5)$). So among small claw-free graphs, $\chi > \omega$ is much more frequent outside the class of line graphs (8.7% against 0.7%).

In set A the gap is positive exactly for class II roots, as (1) predicts. All Erdős–Rényi, bipartite and tree roots were of class I, which agrees with [11, 19]. Co-bipartite graphs are perfect, whereas co-triangle-free graphs have the largest gaps (mean 4.2, maximum 11, χ/ω up to 1.79). As expected, D_N is high in all sets.

6.3 Line graphs and non-line claw-free graphs (RQ1)

Table 3 and Figure 3 compare DSATUR and the V-EDGE model on all sets. When the greedy DSATUR coloring already uses $\omega(G)$ colors, the instance is solved without search. This happened for almost all line graphs of Erdős–Rényi, bipartite and tree roots and for 96% of the co-bipartite graphs. When $\chi > \omega$, the branch and bound has to prove a lower bound, and the search tree grows very quickly. DSATUR reached the time limit on 173 line graphs (165 of them with class II roots) and on half of the co-triangle-free graphs. The V-EDGE model is slower on easy instances because of a fixed overhead of a few milliseconds. With the tested implementations, however, it was clearly more reliable than our Python DSATUR on instances with $\chi > \omega$: it solved all graphs of set A and 80% of the co-triangle-free graphs.

To compare graphs of similar size we restricted set A to $20 \leq n \leq 50$ (362 graphs). In this range DSATUR solved 91.7% of the line graphs and 75.0% of the graphs of set B, and V-EDGE solved 100% and 90.0%. The median of T_V was 7.3 ms against 27.6 ms (Mann–Whitney test, $p < 10^{-40}$). The difference comes from the co-triangle-free family. Its gap can be large, so the vertex models need on average 4.8 solver calls per graph, most of them infeasible. None of the complements of maximal triangle-free graphs on 50 vertices was solved by either method, although their chromatic number follows from a maximum matching. The two sets also differ in density (0.19 against 0.77 in this range), so the result describes the tested families only.

Table 3: Exact DSATUR and the V-EDGE model (time limit 30 s). Medians include runs stopped at the time limit; † at least half of the runs were stopped, so the median is a lower bound.

		Exact DSATUR				V-EDGE model		
Set	Family	Solved [%]	Median [ms]	Max [s]	Median nodes	Solved [%]	Median [ms]	Max [s]
A	Erdős–Rényi	100.0	0.47	0.02	0	100.0	17.0	1.17
A	bipartite	100.0	0.51	0.02	0	100.0	13.1	0.61
A	trees	100.0	0.11	0.00	0	100.0	5.6	0.01
A	regular	82.6	1.82	30.00	214	100.0	14.1	0.34
A	subdivided regular	22.2	30001†	30.00	5055488	100.0	165	22.33
B	co-bipartite	100.0	0.25	0.01	0	100.0	18.7	0.05
B	co-triangle-free	50.0	16389†	30.00	1904076	80.0	103	30.05
C	line	100.0	0.03	0.00	0	100.0	6.1	0.03
C	non-line	100.0	0.03	0.01	0	100.0	6.2	0.02

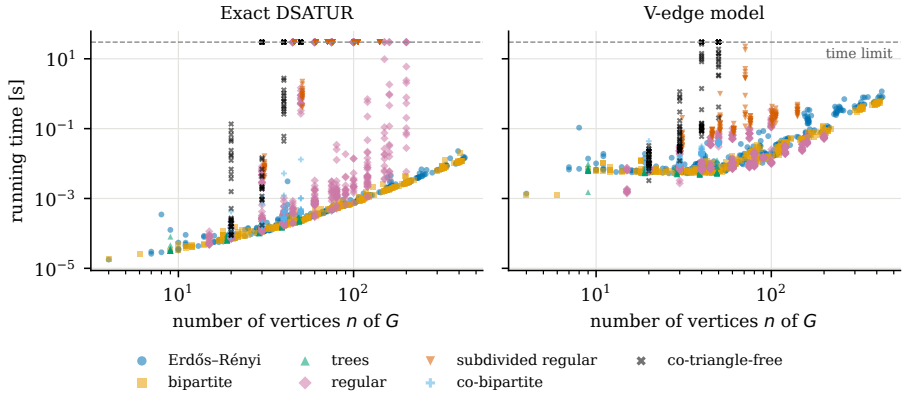


Figure 3: Running time of exact coloring against the number of vertices.

Table 4: Line graphs of set A: median times of the three models and of the root reconstruction, and the speed-ups S and S_{tot} (median), by family of the root and by $R = |E(H)|/|V(H)|$.

		Median time [ms]				$S = T_V/T_E$				
Group	Graphs	T_V	T_C	T_E	T_{rec}	Median	Geo. mean	Max	$S > 1$ [%]	S_{tot}
Erdős–Rényi	285	17.0	10.3	8.1	3.4	1.99	2.41	12.4	95	1.44
bipartite	240	13.1	8.0	7.5	3.0	1.81	1.98	7.1	95	1.27
trees	75	5.6	5.2	5.1	1.0	1.12	1.11	1.9	84	0.93
regular	390	14.1	9.8	7.7	2.3	1.58	1.62	6.0	94	1.23
subdivided regular	135	165	216	134	2.9	1.25	1.24	100.1	70	1.22
$R \le 1.5$	299	6.3	5.7	5.3	1.0	1.16	1.17	7.8	88	0.96
$1.5 < R \le 3$	451	15.1	9.6	8.4	2.2	1.59	1.50	100.1	90	1.22
$3 < R \le 6$	285	52.3	25.1	17.9	5.2	2.61	2.28	7.0	93	1.82
$R > 6$	90	428	165	69.0	28.9	6.60	6.77	12.4	100	4.51
All line graphs	1125	16.7	10.3	8.5	2.5	1.60	1.77	100.1	91	1.23

6.4 Vertex coloring of $L(H)$ and edge coloring of H (RQ2)

All three models solved all 1125 line graphs, so the comparison in Table 4 is complete. E-STAR was faster than V-EDGE on 91% of the graphs (Wilcoxon test on logarithms of the times, $p < 10^{-100}$), which supports H1. The typical gain is moderate: the median of S is 1.60 (quartiles 1.17 and 2.53). If the reconstruction time is added, the root approach is faster on 71% of the graphs and the median of S_{tot} is 1.23 (quartiles 0.98 and 1.80). For the whole set the three models needed 145 s, 143 s and 69 s, and all reconstructions took 6.8 s.

The gain depends on the root graph (Figure 4). S is positively correlated with R ($\rho = 0.71$)

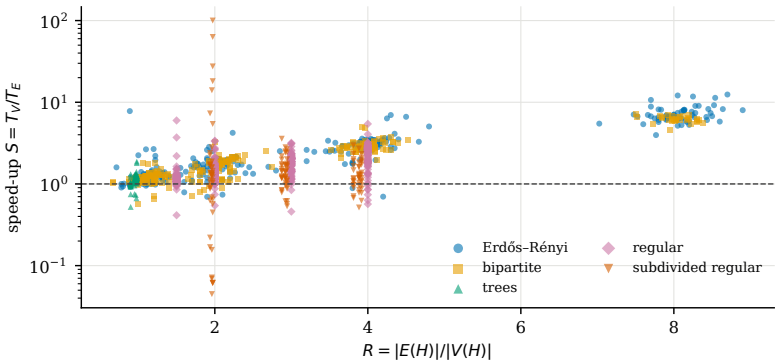


Figure 4: Speed-up $S = T_V/T_E$ against $R = |E(H)|/|V(H)|$ for the line graphs of set A (logarithmic vertical axis).

and with $\Delta(H)$ ($\rho = 0.70$), as expected in H2. The relation holds also within the families ($\rho = 0.89$ for bipartite and 0.88 for Erdős–Rényi roots) and between the 75 parameter combinations of set A ($\rho = 0.82$ for the medians of S and R). For $R \leq 1.5$ the median of S is 1.16 and that of S_{tot} is 0.96; for $R > 6$ they are 6.60 and 4.51, and E-STAR is faster on every instance. The sizes of the models explain this: the median numbers of constraint sets $|\mathcal{Q}|$ are 240 for V-EDGE, 40 for V-CLIQUE and 26 for E-STAR, and the difference grows quadratically with the degrees of H. For sparse roots, such as trees, the models have similar size, and the reconstruction does not pay off ($S_{\text{tot}} = 0.93$).

The V-CLIQUE model shows where the gain comes from. Clique constraints computed from G alone remove most of the difference: the median of T_C/T_E is 1.17, and E-STAR is faster than V-CLIQUE on 85% of the graphs. The remaining advantage of the root graph is that the Krausz partition is a small family of cliques covering all edges, and that Vizing's theorem saves the second solver call for class II roots. In practice the reconstruction is a fast way to obtain a compact clique formulation.

Subdivided regular graphs behave differently. Their times vary by two orders of magnitude between instances of the same size, V-CLIQUE is often the slowest model, and S ranges from 0.04 to 100. Ten of the twelve instances with $S < 0.5$ belong to this family, and within it S is not correlated with R. The decisive step is the infeasibility proof for $k = \Delta$, and its length seems to depend more on the branching of the solver than on the formulation.

6.5 Class I and class II roots

Set A contains 899 line graphs of class I roots and 226 of class II roots: 90 regular graphs of odd order (overfull), the 135 subdivided regular graphs of Observation 1, and one cubic graph on ten vertices that is neither. Table 5 shows that, within the tested families, the class of the root has a large effect. In the regular and subdivided regular families, where orders and degrees are comparable, the median of T_V rises from 9.9 ms for class I to 94.8 ms for class II, and the median of T_E from 6.8 ms to 68.4 ms (Mann–Whitney test, $p < 10^{-70}$ in both cases). For DSATUR the effect is much stronger: it solved 99.1% of the class I instances but only 27.0% of the class II ones.

The overfull instances were easier for the integer programming models than the subdivided graphs, which are not overfull. For every pair (n_0, d) with $n_0 \in \{15, 25\}$, the median times of V-EDGE and E-STAR on the regular graphs were 1.4 to 2.8 times smaller than on the corresponding subdivided graphs. The hardest group, subdivided 4-regular graphs with $n_0 = 35$, had median times of 2.7 s for V-EDGE and 0.52 s for E-STAR, with maxima of 22.3 s and 7.6 s.

6.6 Structural parameters and difficulty (RQ3)

Table 6 gives Spearman's correlations between structural parameters and three measures of difficulty. The size of the DSATUR search tree is only weakly related to the size of the graph ($\rho \approx 0.2$ for n and m), but it is strongly associated with the chromatic gap under the implemented search strategy:

Table 5: Line graphs of set A by the class of the root graph.

			Exact DSATUR		Median model time [ms]			
Class of H	Family	Graphs	Solved [%]	Median nodes	T_V	T_C	T_E	Overfull [%]
I	Erdős–Rényi	285	100.0	0	17.0	10.3	8.1	0
I	bipartite	240	100.0	0	13.1	8.0	7.5	0
I	trees	75	100.0	0	5.6	5.2	5.1	0
I	regular	299	97.3	106	9.9	8.1	6.8	0
I	all families	899	99.1	0	10.3	8.0	6.8	0
II	regular	91	34.1	5006080	55.6	58.8	36.0	99
II	subdivided regular	135	22.2	5055488	165	216	134	0
II	all families	226	27.0	5054592	94.8	121	68.4	40

Table 6: Spearman's ρ between structural parameters and measures of difficulty. T_V on set B is restricted to solved instances; node counts of runs stopped at the time limit are the counts at interruption. Values marked with $^\circ$ are not significant at the level 0.001.

	Set A (line graphs)			Set B (non-line)	
Parameter	T_V	T_E	DSATUR nodes	T_V	DSATUR nodes
n	+0.75	+0.63	+0.16	+0.67	+0.20°
m	+0.82	+0.68	+0.19	+0.68	+0.21°
density	−0.13	−0.10°	−0.01°	+0.35	+0.10°
$\Delta(G)$	+0.66	+0.49	−0.11	+0.63	+0.13°
$\omega(G)$	+0.62	+0.45	−0.18	+0.41	−0.23
$\chi - \omega$	+0.50	+0.61	+0.77	+0.66	+0.95
degeneracy	+0.76	+0.61	+0.20	+0.70	+0.25
$D_N(G)$	+0.35	+0.21	−0.20	+0.28	−0.03°

$\rho = 0.77$ on set A and $\rho = 0.95$ on set B. This is partly a consequence of the algorithm itself. If the heuristic finds an ω-coloring, the search stops at once; if $\chi > \omega$, the lower bound ω can never close the gap. The association remains within single families ($\rho = 0.68$ for regular roots, 0.76 for co-triangle-free graphs). The running times of the integer programming models are related mainly to size: for T_V the strongest correlates are m ($\rho = 0.82$), the degeneracy (0.76) and n (0.75), and the gap follows with 0.50. For T_E the gap (0.61) is as strongly associated as the degeneracy (0.61) and n (0.63), because the compact model removes much of the effect of size. Density and D_N show weak associations in all cases.

These parameters are themselves correlated (for example, the degeneracy grows with m), so a single coefficient does not show which of them causes the difficulty. H3 is therefore confirmed only in part. The gap is clearly more closely related than n to the work of branch and bound, the degeneracy is about as good as n and m for the models, and D_N is not informative in a class where it is always high. The gap is also known only after the problem has been solved. For line graphs it is determined by the class of the root, and only sufficient conditions, such as overfullness, are cheap to test.

7 Discussion

The answer to RQ2 is positive, with reservations. With the reconstruction included, the root approach was faster on 71% of the line graphs and on all graphs with dense roots, while for sparse roots it gave no gain. The conflict structure of $L(H)$ is then described by $|V(H)|$ stars instead of $\sum_v \binom{d_H(v)}{2}$ edges. The V-CLIQUE model shows that most of this gain is the known advantage of clique constraints over edge constraints. What the root graph adds is a compact clique cover obtained in linear time and the restriction of the search to one value of k. It does not make hard instances easy: proving that a class II graph has no Δ-edge-coloring is the most expensive step in

every formulation.

For RQ1, the experiments suggest that the difference between easy and hard instances depends less on whether a graph is a line graph than on whether $\chi = \omega$. In line graphs Vizing's theorem keeps the gap at most one. In complements of triangle-free graphs the gap can be large, and both general methods failed on some of them at 50 vertices, although a matching algorithm colors them optimally. In both parts of the class, knowledge of the structure helps, but the relevant structure is different.

Limitations. (i) Only one integer programming solver with default settings was used; a constraint programming solver with `AllDifferent` propagation, such as CP-SAT [25], may rank the models differently. The software contains such a back-end, but it was not used in the reported experiments. (ii) DSATUR is written in pure Python and lacks the improvements of [14, 27], so its times are pessimistic and the comparison with HiGHS concerns these implementations only. (iii) Most instances are solved within milliseconds, where timing noise is not negligible. (iv) Set B consists of two dense families with $\alpha \leq 2$; in particular, it contains no quasi-line graphs that are not line graphs. (v) Specialized exact solvers [12, 16, 30] are much stronger than our baselines on general benchmarks and should be included in further work.

8 Conclusions

We compared vertex coloring of line graphs with edge coloring of their reconstructed roots, using DSATUR and three integer programming models on three test sets of claw-free graphs. The main findings are:

1. The root model was faster than the edge model on G for 91% of 1125 line graphs (median speed-up 1.60); with the reconstruction time included, for 71% (median 1.23). The gain grows with $|E(H)|/|V(H)|$ and reaches a median of 6.6 for the densest roots (H1 and H2 supported).
2. Most of the advantage is already obtained with clique constraints computed from G (median times 16.7, 10.3 and 8.5 ms for V-EDGE, V-CLIQUE and E-STAR). The root graph adds a compact clique cover and the two-valued bound of Vizing's theorem.
3. Within the tested families, line graphs of class II roots were about ten times slower for the integer programming models, and DSATUR solved only 27% of them.
4. Among the tested claw-free graphs that are not line graphs, co-bipartite graphs were easy, and the co-triangle-free family contained the hardest instances of the study.
5. Under the implemented search strategy, the chromatic gap is the parameter most strongly associated with the size of the DSATUR search tree, while the times of the integer programming models follow the size and degeneracy of the graph (H3 supported in part).

Further work may include a constraint programming solver and current SAT-based coloring solvers [12, 16], cheap structural tests (König's theorem, overfull subgraphs) in the root model, quasi-line graphs and claw-free graphs with $\alpha \geq 3$, and a regression model that predicts the running time from parameters known before solving.

Data availability. The source code, the instance generators with their seeds and the raw results are available at [repository URL].

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